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EXPT = 9

TITLE = Prim's Algorithm

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In [1]: 1
2 n = int(input("Enter number of vertices in the graph: "))
3
4 print("Enter the adjacency matrix where each row separated by spaces:")
5 graph = []
6 for i in range(n):
7     row = input().split()
8     graph.append([int(x) for x in row])
9
10 def get_min_key_vertex(key, in_mst):
11     return min((v for v in range(len(key)) if not in_mst[v]), key=lambda v: key[v], default=-1)
12
13 def Prim_MCST(graph):
14     vertices = len(graph)
15     parent = [-1] * vertices
16     key = [float('inf')] * vertices
17     in_mst = [False] * vertices
18     key[0] = 0
19     total_cost = 0
20
21     for _ in range(vertices):
22         u = min((v for v in range(vertices) if not in_mst[v]), key=lambda v: key[v], default=-1)
23         if u == -1:
24             break
25         in_mst[u] = True
26
27         for v in range(vertices):
28             if graph[u][v] != 0 and not in_mst[v] and graph[u][v] < key[v]:
29                 key[v] = graph[u][v]
30                 parent[v] = u
31
32     total_cost = sum(key[1:])
33
34     return total_cost
35
36 print("=====")
37 min_cost = Prim_MCST(graph)
38 print("Minimum Cost of Spanning Tree using Prims Algorithm: ", min_cost)
39 print("=====")
```

```
Enter number of vertices in the graph: 7
Enter the adjacency matrix where each row separated by spaces:
0 28 0 0 0 10 0
28 0 16 0 0 0 14
0 16 0 12 0 0 0
0 0 12 0 22 0 18
0 0 0 22 0 25 24
10 0 0 0 25 0 0
0 14 0 18 24 0 0
=====
Minimum Cost of Spanning Tree using Prims Algorithm: 99
=====
```